PR2 plan preview — Gantt table

ID	Task	Status	Start	End	Dependencies
1	Iteration 1: PR1 Planning and Coordination	In Progress	Oct 10	Oct 22	
1.01	Confirm project scope and final service list	In Progress	Oct 10	Oct 14	
1.02	Confirm microservice owners and responsibilities	Complete	Oct 10	Oct 14	1.01
1.03	Create task list	Complete	Oct 10	Oct 14	1.01,1.02
1.04	Create Gantt chart and iteration schedule	Complete	Oct 10	Oct 14	1.03
1.05	Review planning and documents for consistency	Complete	Oct 14	Oct 14	1.03,1.04
1.06	Draft PR1 written documentation and speadsheets	Complete	Oct 14	Oct 14	1.05
1.07	Draft PR1 slide outline and assign slide sections	Complete	Oct 14	Oct 14	1.05
1.08	Make slide deck	Complete	Oct 14		1.07
1.09	Review slides against PR1 deliverables	Complete	Oct 21	Oct 22	1.08
1.1	Practice presentation	Complete	Oct 21	Oct 22	1.09
2	Iteration 2: Requirements and Service Definition	Complete	Oct 10	Oct 14	
2.01	Define services, IDs, titles, and descriptions	Complete	Oct 10	Oct 14	1.01
2.02	Define functional requirements for each service	Complete	Oct 10	Oct 14	2.01
2.03	Define quality requirements	Complete	Oct 10	Oct 14	2.01
2.04	Identify potential ASRs and prepare justifications	Complete	Oct 10	Oct 14	2.02,2.03
3	Iteration 3: Cloud and Architecture Description	In Progress	Oct 10	Nov 11	
3.01	Describe cloud platform familiarity	Complete	Oct 14	Oct 14	1.01
3.02	Draft cloud deployment approach	Complete	Oct 10	Oct 14	1.01
3.03	Describe microservice architecture familiarity	Complete	Oct 14	Oct 14	1.01
3.04	Preliminary architecture sketch	In Progress	Oct 21	Nov 11	3.02,3.03
4	Iteration 4: Implementation and Integration	In Progress	Oct 21	Nov 11	
4.01	Set up cloud environment and repository	In Progress	Oct 21	Nov 11	
4.02	Implement service registry/discovery	In Progress	Oct 21	Nov 11	4.01
4.03	Implement service consumer GUI	Not Started	Oct 21	Nov 11	4.02,4.09
4.04	Integrate Alpha Vantage data source and confirm access	In Progress	Oct 21	Nov 11	4.01
4.05	Retrieve data for required one-year period and selected companies	Not Started	Oct 21	Nov 11	4.04
4.06	Handle API limits, failed requests, missing data, and invalid responses	In Progress	Oct 21	Nov 11	4.04
4.07	Decide and implement stock-data storage, caching, or retrieval	In Progress	Oct 21	Nov 11	4.04
4.08	Verify stock data completeness and correctness	In Progress	Oct 21	Nov 11	4.05,4.06
4.09	Define HTTP endpoints and request/response formats for each service	Not Started	Oct 21	Nov 11	2.01,2.02
5	Iteration 5: Microservice Implementation	In Progress	Oct 21	Nov 11	
5.01	Implement microservice 1	Not Started	Oct 21	Nov 11	
5.02	Implement microservice 2	Complete	Oct 21	Nov 11	
5.03	Implement microservice 3	In Progress	Oct 21	Nov 11	

ID	Task	Status	Start	End	Dependencies
5.04	Implement microservice 4	In Progress	Oct 21	Nov 11	4.04,4.09
5.05	Implement microservice 5	In Progress	Oct 21	Nov 11	4.04,4.09
5.06	Implement consistent validation and error responses across services	Not Started	Oct 21	Nov 11	
5.07	Test each microservice independently with representative requests	Not Started	Oct 21	Nov 11	
5.08	Verify each service can be deployed and accessed independently	Not Started	Oct 21	Nov 11	
6	Iteration 6: Integration, Reliability, Documentation, and Demo	In Progress	Oct 21	Nov 12	
6.01	Implement dynamic registration and removal	Not Started	Oct 21		
6.02	Implement backup registry and failover	Not Started	Oct 21	Nov 12	
6.03	Create architecture views and explanations	Not Started	Oct 21	Nov 12	3.04,6.18,6.19,6.20
6.04	Integrate services and GUI	Not Started	Oct 21	Nov 11	4.03,4.09
6.05	Test discovery, removal, and registry failure	Not Started	Oct 21		
6.06	Prepare preliminary demo	Complete	Oct 21	Nov 12	
6.07	Test registration, deregistration, and registry search APIs	Not Started	Oct 21		
6.08	Verify search results show service names, descriptions, and URLs	Not Started	Oct 21		
6.09	Test clicking results and navigating to the correct service page	Not Started	Oct 21		
6.1	Verify removed service is no longer listed or accessible through intended access path	Not Started	Oct 21		
6.11	Simulate primary registry failure and verify automatic backup redirection	Complete	Oct 21		
6.12	Test registry recovery and consistency after primary returns	Not Started	Oct 21		
6.13	Deploy registry, GUI, and microservices to selected cloud platform	Complete	Oct 21		
6.14	Verify deployed application end-to-end from consumer GUI	Not Started	Oct 21		
6.15	Test adding/removing a service while rest of system remains running	Not Started	Oct 21		
6.16	Run end-to-end acceptance tests against project requirements	In Progress	Oct 21		
6.17	Complete service catalogue and requirements/ASR justifications	Not Started	Oct 21	Nov 12	2.01,2.02,2.03,2.04
6.18	Document Module view	Complete	Oct 21	Nov 11	3.04
6.19	Document Component-and-Connector view	Complete	Oct 21	Nov 11	3.04
6.2	Document three quality views (e.g. reliability, communication, error handling)	In Progress	Oct 21	Nov 11	6.18,6.19
6.21	Prepare demo instructions and verify demo in target environment	In Progress	Oct 21	Nov 12	6.27
6.22	Complete AI comparison table and lessons learned where AI was used	In Progress	Oct 21	Nov 12	
6.23	Combined Architecture View	In Progress	Oct 21	Nov 11	6.18,6.19
6.24	PR2 Presentation Slides	In Progress	Oct 21	Nov 11	6.17,6.18,6.19,6.20
6.25	PR2 Presentation Rehearsal	Not Started	Oct 21	Nov 12	6.24,6.21
6.26	PR2 Binder and Submission	Not Started	Oct 21	Nov 12	6.24,6.25
6.27	PR2 Integration Version	Not Started	Oct 21	Nov 11	6.04,5.07

Service task list — all owners retained

Row / Gantt	Owner	Task	Status / due	Notes
3 / 4.01	Aashka	Configure the selected cloud environment and verify access	Not Started / Nov 11	Confirm platform and deployment approach first
4 / 4.01	Emily	Create repository structure and establish branch/PR workflow	Complete / Nov 11	Coordinate with cloud setup
5 / 4.02	Aashka	Define the service registration record and registry API	Not Started / Nov 11	Agree on service metadata and communication protocol
6 / 4.02	Aashka	Implement service registration and update	Not Started / Nov 11	Registry API defined
7 / 6.01	Aashka	Implement service deregistration when a service shuts down	Not Started / Nov 11	Registration implemented
8 / 4.02	Aashka	Implement service lookup/search and return service name, description, and URL	Complete / Nov 11	Registration records defined
9 / 4.03	Gordon	Create search interface for discovering available services	Not Started / Nov 11	Registry lookup API defined
10 / 4.03	Gordon	Display clickable service results and navigate to a service URL	Not Started / Nov 11	Search interface and service URL format defined
11 / 6.02	Aashka	Configure or implement a backup registry	Not Started / TBD	Primary registry design and deployment approach
12 / 6.02	Aashka	Implement automated redirection/failover to the backup registry	Not Started / TBD	Backup registry available
13 / 5.01	Gordon	Confirm service purpose, inputs, outputs, and API contract	Not Started / Nov 11	Finalize service catalogue
14 / 5.01	Gordon	Implement core service logic	Not Started / Nov 11	API contract agreed
15 / 5.01	Gordon	Connect service to required stock/company data source	Not Started / Nov 11	Confirm Alpha Vantage data requirements and API limits
16 / 5.01	Gordon	Add input validation and error responses	Not Started / Nov 11	Core endpoint implemented
17 / 5.01	Gordon	Register service at startup and deregister on shutdown	Not Started / Nov 11	Registry API available
18 / 5.01	Gordon	Test normal and error cases	Not Started / Nov 11	Core logic and validation implemented
19 / 4.04	Gordon	Integrate Alpha Vantage data source and confirm access	Not Started / Nov 11	Confirm platform and API access | Scope: S1. These are per-service tasks, not repeated shared completion claims.
20 / 4.05	Gordon	Retrieve data for the required one-year period and selected companies	Not Started / Nov 11	Confirm symbols and exact date range | Scope: S1. These are per-service tasks, not repeated shared completion claims.
21 / 4.06	Gordon	Handle API limits, failed requests, missing data, and invalid responses	Not Started / Nov 11	Include meaningful error handling | Scope: S1. These are per-service tasks, not repeated shared completion claims.
22 / 4.07	Gordon	Decide and implement stock data storage, caching, or retrieval	Not Started / Nov 11	Architecture decision | Scope: S1. These are per-service tasks, not repeated shared completion claims.
23 / 4.08	Gordon	Verify stock data completeness and correctness	Not Started / Nov 11	Test required companies and dates | Scope: S1. These are per-service tasks, not repeated shared completion claims.
24 / 5.02	Max	Confirm service purpose, inputs, outputs, and API contract	Complete / TBD	Finalize service catalogue

Row / Gantt	Owner	Task	Status / due	Notes
25 / 5.02	Max	Implement core service logic	Complete / TBD	API contract agreed
26 / 5.02	Max	Connect service to required stock/company data source	Complete / TBD	Confirm Alpha Vantage data requirements and API limits
27 / 5.02	Max	Add input validation and error responses	Complete / TBD	Core endpoint implemented
28 / 5.02	Max	Register service at startup and deregister on shutdown	Complete / TBD	Registry API available
29 / 5.02	Max	Test normal and error cases	Complete / TBD	Core logic and validation implemented
30 / 4.04	Max	Integrate Alpha Vantage data source and confirm access	Not Started / Nov 11	Confirm platform and API access | Scope: S2. These are per-service tasks, not repeated shared completion claims.
31 / 4.05	Max	Retrieve data for the required one-year period and selected companies	Not Started / Nov 11	Confirm symbols and exact date range | Scope: S2. These are per-service tasks, not repeated shared completion claims.
32 / 4.06	Max	Handle API limits, failed requests, missing data, and invalid responses	Not Started / Nov 11	Include meaningful error handling | Scope: S2. These are per-service tasks, not repeated shared completion claims.
33 / 4.07	Max	Decide and implement stock data storage, caching, or retrieval	Not Started / Nov 11	Architecture decision | Scope: S2. These are per-service tasks, not repeated shared completion claims.
34 / 4.08	Max	Verify stock data completeness and correctness	Not Started / Nov 11	Test required companies and dates | Scope: S2. These are per-service tasks, not repeated shared completion claims.
35 / 5.03	Gordon	Confirm service purpose, inputs, outputs, and API contract	Not Started / TBD	Finalize service catalogue
36 / 5.03	Aashka	Implement core service logic	Complete / TBD	API contract agreed
37 / 5.03	Aashka	Connect service to required stock/company data source	Complete / TBD	Confirm Alpha Vantage data requirements and API limits
38 / 5.03	Aashka	Add input validation and error responses	Not Started / TBD	Core endpoint implemented
39 / 5.03	Aashka	Register service at startup and deregister on shutdown	Not Started / TBD	Registry API available
40 / 5.03	Aashka	Test normal and error cases	Not Started / TBD	Core logic and validation implemented
41 / 4.04	Aashka	Integrate Alpha Vantage data source and confirm access	Not Started / Nov 11	Confirm platform and API access | Scope: S3. These are per-service tasks, not repeated shared completion claims.
42 / 4.05	Aashka	Retrieve data for the required one-year period and selected companies	Not Started / Nov 11	Confirm symbols and exact date range | Scope: S3. These are per-service tasks, not repeated shared completion claims.
43 / 4.06	Aashka	Handle API limits, failed requests, missing data, and invalid responses	Not Started / Nov 11	Include meaningful error handling | Scope: S3. These are per-service tasks, not repeated shared completion claims.
44 / 4.07	Aashka	Decide and implement stock data storage, caching, or retrieval	Not Started / Nov 11	Architecture decision | Scope: S3. These are per-service tasks, not repeated shared completion claims.
45 / 4.08	Aashka	Verify stock data completeness and correctness	Not Started / Nov 11	Test required companies and dates | Scope: S3. These are per-service tasks, not repeated shared completion claims.
46 / 5.04	Gordon	Confirm service purpose, inputs, outputs, and API contract	Not Started / TBD	Finalize service catalogue
47 / 5.04	Emily	Implement core service logic	Complete / Nov 11	API contract agreed
48 / 5.04	Emily	Connect service to required stock/company data source	Complete / Nov 11	Confirm Alpha Vantage data requirements and API limits
49 / 5.04	Emily	Add input validation and error responses	Complete / Nov 11	Core endpoint implemented
50 / 5.04	Emily	Register service at startup and deregister on shutdown	In Progress / Nov 11	Registry API available | Implemented/tested registration, heartbeat recovery and graceful lease cessation. Local test: S4 expired after 11.41 s while S5 remained healthy. Immediate deregistration endpoint absent in registry; owner contract remains pending.

Row / Gantt	Owner	Task	Status / due	Notes
51 / 5.04	Emily	Test normal and error cases	Complete / Nov 11	Core logic and validation implemented | Current suites: S4 56 and S5 61 offline tests; additional real local registry HTTP test passed. Cloud/GUI and authenticated provider testing remain separate.
52 / 4.04	Emily	Integrate Alpha Vantage data source and confirm access	In Progress / Nov 11	Confirm platform and API access | Request-driven provider integration and mocked tests complete. User confirms no Alpha Vantage key configured; authenticated access not verified. | Scope: S4. These are per-service tasks, not repeated shared completion claims.
53 / 4.05	Emily	Retrieve data for the required one-year period and selected companies	Not Started / Nov 11	Confirm symbols and exact date range | Blocked by provider key/entitlement. Symbols and dates remain user-selected at request time; no fixed company list or actual one-year dataset is claimed. | Scope: S4. These are per-service tasks, not repeated shared completion claims.
54 / 4.06	Emily	Handle API limits, failed requests, missing data, and invalid responses	Complete / Nov 11	Include meaningful error handling | S4/S5 source tests verify limits, bad payloads, invalid/missing observations, timeout errors and recovery. Offline implementation evidence; no live-provider recovery claim. | Scope: S4. These are per-service tasks, not repeated shared completion claims.
55 / 4.07	Emily	Decide and implement stock data storage, caching, or retrieval	Complete / Nov 11	Architecture decision | Implemented request-driven retrieval, no mandatory shared database or S1 dependency, no synthetic fallback; decision recorded in docs/data-retrieval-decision.md. | Scope: S4. These are per-service tasks, not repeated shared completion claims.
56 / 4.08	Emily	Verify stock data completeness and correctness	In Progress / Nov 11	Test required companies and dates | Fixed-data numerical/coverage checks pass. Actual one-year source completeness and correctness await an entitled key and a live request. | Scope: S4. These are per-service tasks, not repeated shared completion claims.
57 / 5.05	Max	Confirm service purpose, inputs, outputs, and API contract	Not Started / TBD	Finalize service catalogue
58 / 5.05	Emily	Implement core service logic	Complete / Nov 11	API contract agreed
59 / 5.05	Emily	Connect service to required stock/company data source	Complete / Nov 11	Confirm Alpha Vantage data requirements and API limits
60 / 5.05	Emily	Add input validation and error responses	Complete / Nov 11	Core endpoint implemented
61 / 5.05	Emily	Register service at startup and deregister on shutdown	In Progress / Nov 11	Registry API available | Implemented/tested registration, heartbeat recovery and graceful lease cessation. Local test: S4 expired after 11.41 s while S5 remained healthy. Immediate deregistration endpoint absent in registry; owner contract remains pending.
62 / 5.05	Emily	Test normal and error cases	Complete / Nov 11	Core logic and validation implemented | Current suites: S4 56 and S5 61 offline tests; additional real local registry HTTP test passed. Cloud/GUI and authenticated provider testing remain separate.
63 / 4.04	Emily	Integrate Alpha Vantage data source and confirm access	In Progress / Nov 11	Confirm platform and API access | Request-driven provider integration and mocked tests complete. User confirms no Alpha Vantage key configured; authenticated access not verified. | Scope: S5. These are per-service tasks, not repeated shared completion claims.
64 / 4.05	Emily	Retrieve data for the required one-year period and selected companies	Not Started / Nov 11	Confirm symbols and exact date range | Blocked by provider key/entitlement. Symbols and dates remain user-selected at request time; no fixed company list or actual one-year dataset is claimed. | Scope: S5. These are per-service tasks, not repeated shared completion claims.

Row / Gantt	Owner	Task	Status / due	Notes
65 / 4.06	Emily	Handle API limits, failed requests, missing data, and invalid responses	Complete / Nov 11	Include meaningful error handling | S4/S5 source tests verify limits, bad payloads, invalid/missing observations, timeout errors and recovery. Offline implementation evidence; no live-provider recovery claim. | Scope: S5. These are per-service tasks, not repeated shared completion claims.
66 / 4.07	Emily	Decide and implement stock data storage, caching, or retrieval	Complete / Nov 11	Architecture decision | Implemented request-driven retrieval, no mandatory shared database or S1 dependency, no synthetic fallback; decision recorded in docs/data-retrieval-decision.md. | Scope: S5. These are per-service tasks, not repeated shared completion claims.
67 / 4.08	Emily	Verify stock data completeness and correctness	In Progress / Nov 11	Test required companies and dates | Fixed-data numerical/coverage checks pass. Actual one-year source completeness and correctness await an entitled key and a live request. | Scope: S5. These are per-service tasks, not repeated shared completion claims.
68 / 4.09	Max,Gordon	Define HTTP endpoints and request/response formats for each service	Not Started / TBD	Finalize service catalogue
69 / 5.06	Gordon	Implement consistent validation and error responses across services	Not Started / TBD	Apply to all five services
70 / 5.07	Gordon	Test each microservice independently with representative requests	Not Started / TBD	Include successful and unsuccessful requests
71 / 5.08	Gordon	Verify each service can be deployed and accessed independently	Not Started / TBD	Confirm deployment URL
72 / 6.07	Unassigned	Test registration, deregistration, and registry search APIs	Not Started / Nov 11	Verify registry state
73 / 6.08	Unassigned	Verify search results show service names, descriptions, and URLs	Not Started / Nov 11	Required consumer-facing information
74 / 6.09	Unassigned	Test clicking results and navigating to the correct service page	Not Started / Nov 11	Test valid and unavailable URLs
75 / 6.1	Unassigned	Verify removed service is no longer listed or accessible through intended access path	Not Started / TBD	Clarify system-controlled access path
76 / 6.11	Aashka	Simulate primary registry failure and verify automatic backup redirection	Complete / TBD	Demonstrate automated failover
77 / 6.12	Aashka	Test registry recovery and consistency after primary returns	Not Started / TBD	Document recovery behaviour
78 / 6.13	Aashka	Deploy registry, GUI, and microservices to selected cloud platform	Complete / TBD	Verify deployed URLs
79 / 6.14	All	Verify deployed application end-to-end from consumer GUI	Not Started / TBD	Use deployed endpoints
80 / 6.15	All	Test adding/removing a service while the rest of the system remains running	Not Started / TBD	Check other services remain operational
81 / 6.16	All	Run end-to-end acceptance tests against project requirements	In Progress / TBD	Record pass/fail results | Acceptance matrix and known results prepared; final system/cloud acceptance execution remains pending.
82 / 6.17	Max/Gordon	Complete service catalogue and requirements/ASR justifications	Not Started / Nov 11	Trace requirements to implementation and tests
83 / 6.18	Emily	Document Module view	Complete / Nov 12	Use required architecture view template | Delivered PR2 view with primary diagram, element catalogue, context, variability and architecture background; implemented/planned boundaries and source evidence included.

Row / Gantt	Owner	Task	Status / due	Notes
84 / 6.19	Emily	Document Component-and-Connector view	Complete / Nov 12	Show runtime components and communication | Delivered PR2 view with primary diagram, element catalogue, context, variability and architecture background; implemented/planned boundaries and source evidence included.
85 / 6.2	Gordon/Emily/ Gordon	Document three quality views (e.g. reliability, communication, error handling)	In Progress / Nov 12	Use required template for each view | Emily communication view complete. The two other quality views remain Gordon/team work; do not mark all three complete.
86 / 6.21	All	Prepare demo instructions and verify demo in target environment	In Progress / Nov 12	Include fallback plan | Demo runbook and local lifecycle evidence prepared; actual GUI/cloud verification and rehearsal pending.
87 / 6.22	All	Complete AI comparison table and lessons learned where AI was used	In Progress / Nov 12	Own solution, AI-aided solution, comparison, lessons learned | AI comparison/lessons draft prepared; student-authored baseline was not supplied, and no independent authorship claim is invented.
88 / 3.04	Unassigned	Create or update the high-level architecture diagram and explain the registry, consumer GUI, microservices, and their interactions	In Progress / TBD	Service catalogue and communication interfaces defined. Clearly distinguish implemented components from planned components | System overview provided in Emily's architecture packet; unassigned owner/team to review.
89 / 1.03	Emily	Review task assignments, statuses, completed work, and remaining work for PR2	Complete / Nov 11	Confirm progress with each team member. Distinguish offline testing from live-data and deployment verification. | Prepared dated evidence-based planning copy with PR2 milestones, dependencies and status reconciliation. Other owners still confirm their task completion and unassigned work.
90 / 1.04	Emily	Update the Gantt chart with PR2 deadlines, dependencies, and remaining final-project milestones	Complete / Nov 11	Updated task list available. Show PR2 deliverables due November 12 and separate later final-delivery work | Prepared dated evidence-based planning copy with PR2 milestones, dependencies and status reconciliation. Other owners still confirm their task completion and unassigned work.
91 / 6.04	Unassigned	Connect GUI discovery results to actual service endpoints and display their results or structured errors	Not Started / Nov 11	Registry lookup API, GUI, and demonstrated service APIs available. Clearly label simulated data and verify the complete user interaction
92 / 6.23	Unassigned	Identify a useful combination of architecture views and document the combined diagram and its purpose	In Progress / Nov 11	Structural views drafted. For example, combine module decomposition with implementation ownership. Explain why the combination is useful | Combined ownership/decomposition example provided in Emily's packet; team to review.
93 / 6.24	All	Prepare slides covering changes since PR1, architecture views, implementation progress, limitations, and the preliminary demo	In Progress / Nov 11	Updated requirements, plan, architecture documentation, and agreed demo scope available | Editable PR2 support deck prepared; merge Gordon quality views and team updates before calling the full presentation complete.
94 / 6.25	All	Assign presentation and demo sections, rehearse the complete presentation, and check timing and transitions	Not Started / Nov 11	Slides drafted and preliminary demo working. Confirm each member's contribution and prepare a fallback for demo failures | Presenter/run-of-show proposal prepared; human rehearsal and timing have not occurred.
95 / 6.26	Unassigned	Update the title page and table of contents, check file formats and ordering, package the project binder, and submit to OWL	Not Started / Nov 11	PR2 documents, slides, and spreadsheets finalized. Follow the required folder naming and ZIP format; verify the exact submission deadline and successful upload
96 / 6.27	Unassigned	Resolve blocking review findings and assemble one tested integration version containing the components selected for the preliminary demo	Not Started / Nov 11	Required component branches available and reviews addressed. Follow the repository approval workflow, verify compatibility, and record the commit used for the demo